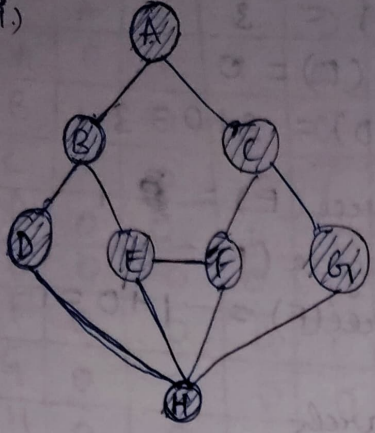


(1)



DFS order:-
A B D H E
F C G

A: [B, C]
B: [D, E]
C: [F, G]
D: [H]
E: [B, F, H]
F: [D, E, H]
G: [C, H]
H: [D, E, F, G]

A	B	C	D	E	F	G	H
0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1

step 1:-

DFS(A)

step 2:-

DFS(B)
DFS(A)

step 3:-

DFS(D)
DFS(B)
DFS(A)

step 4:-

DFS(H)
DFS(D)
DFS(B)
DFS(A)

step 5:-

DFS(E)
DFS(H)
DFS(D)
DFS(B)
DFS(A)

step 6:-

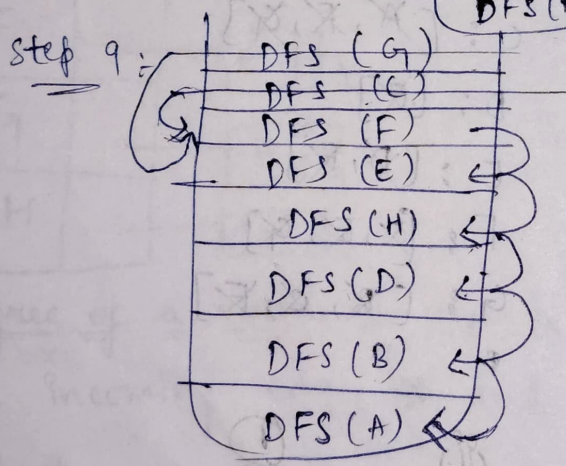
DFS(F)
DFS(E)
DFS(H)
DFS(D)
DFS(B)
DFS(A)

step 7:-

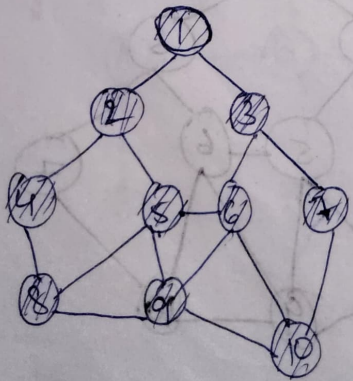
DFS(C)
DFS(F)
DFS(E)
DFS(H)
DFS(D)
DFS(B)
DFS(A)

step 8:-

DFS(G)
DFS(C)
DFS(F)
DFS(E)
DFS(H)
DFS(D)
DFS(B)
DFS(A)



(2)



DFS order:-

1 2 4 8 5 6 3 7 10
9

1	2	3	4	5	6	7	8	9	10
0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1

- 1: [1, 4, 5]
- 2: [1, 6, 7]
- 3: [2, 8]
- 4: [2, 6, 8, 9]
- 5: [3, 5, 9, 10]
- 6: [3, 5, 9, 10]
- 7: [3, 10]
- 8: [4, 8, 9]
- 9: [5, 6, 8, 10]
- 10: [6, 7, 9]

Step 1:

DFS(1)

Step 2:

DFS(2)
DFS(1)

Step 3:

DFS(4)
DFS(2)
DFS(1)

Step 4 & 5:

DFS(5)
DFS(8)
DFS(4)
DFS(2)
DFS(1)

Step 6:

DFS(5)
DFS(8)
DFS(4)
DFS(2)
DFS(1)

Step 7:

DFS(3)
DFS(6)
DFS(5)
DFS(8)
DFS(4)
DFS(2)
DFS(1)

Step 8 & 9:

DFS(7)
DFS(3)
DFS(6)
DFS(5)
DFS(8)
DFS(4)
DFS(2)
DFS(1)

Step 10:

DFS(10)
...
DFS(1)

Step 11:

DFS(9)
DFS(10)
...
DFS(1)

Step 12:

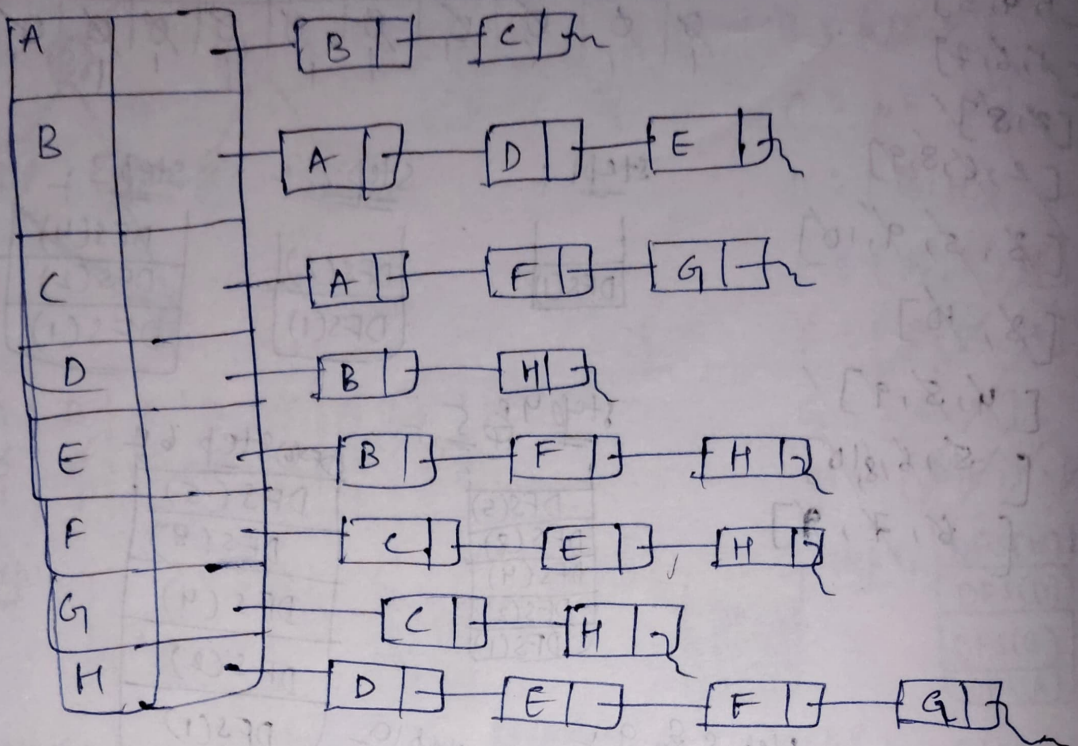
DFS(9)
DFS(10)
DFS(7)
DFS(3)
DFS(6)
DFS(5)
DFS(8)
DFS(4)
DFS(2)
DFS(1)

(P)

	A	B	C	D	E	F	G	H
A	0	1	1	0	0	0	0	0
B	1	0	0	1	1	0	0	0
C	1	0	0	0	0	1	1	0
D	0	1	0	0	0	0	0	1
E	0	1	0	0	0	1	0	1
F	0	0	1	0	1	0	0	1
G	0	0	1	0	0	0	0	1
H	0	0	0	1	1	1	1	0

Adjacency matrix

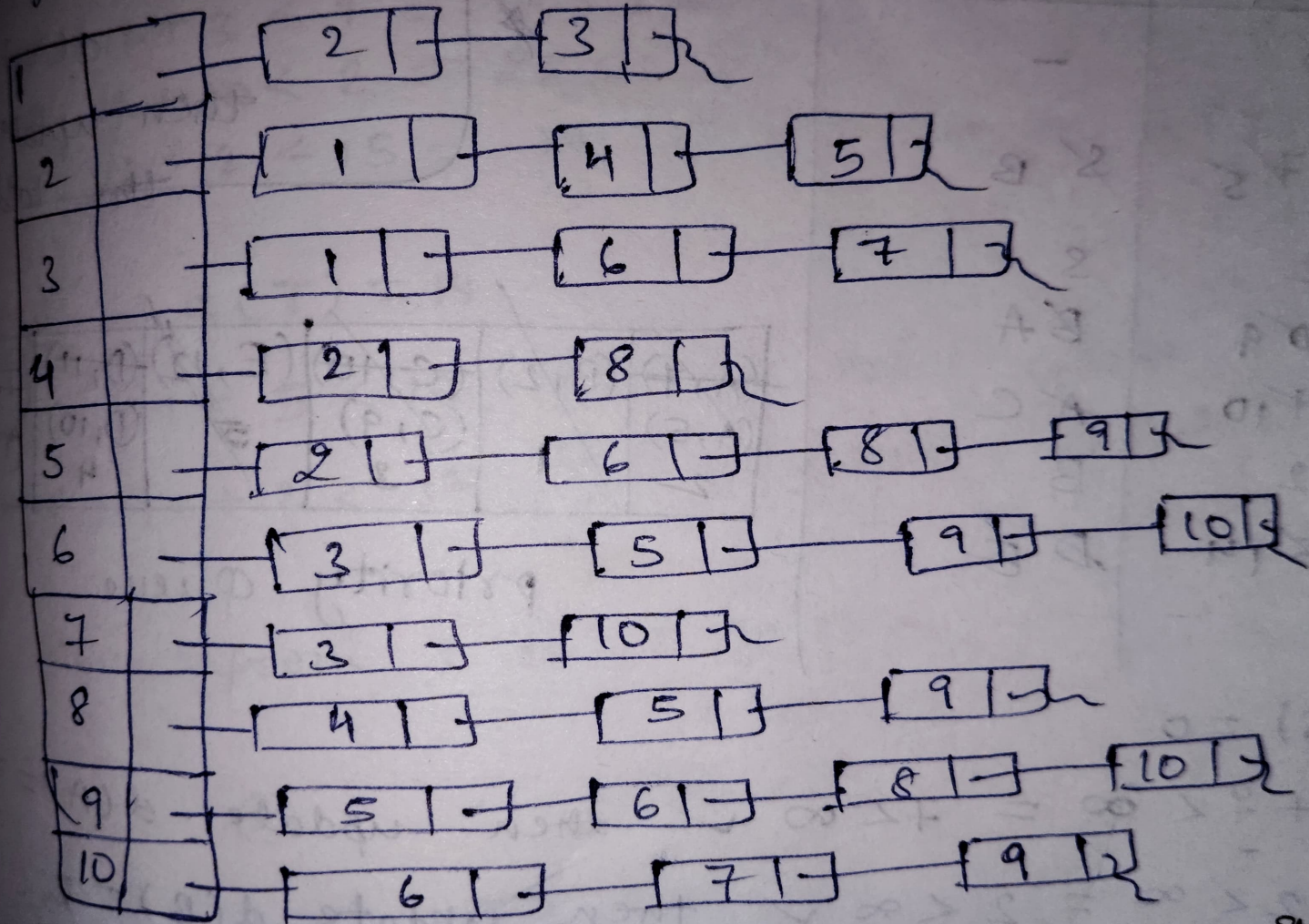
Adjacency list:-



8. Adjacency matrix:-

	1	2	3	4	5	6	7	8	9	10
1	0	1	1	0	0	0	0	0	0	0
2	1	0	0	1	1	0	0	0	0	0
3	1	0	0	0	0	1	1	0	0	0
4	0	1	0	0	0	0	0	1	0	0
5	0	1	0	0	0	1	0	1	1	0
6	0	0	1	0	1	0	0	0	1	1
7	0	0	1	0	0	0	0	0	0	1
8	0	0	0	1	1	0	0	0	1	0
9	0	0	0	0	1	1	0	1	0	1
10	0	0	0	0	0	1	1	0	1	0

Adjacency List :-



Mr. S

(Kruskal's Algorithm)